



KrishiOS: A Sustainable IoT-Based Agriculture and Aquaculture System Using Solar Energy and Intelligent Resource Optimization

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Abstract—One-third of global energy feeds food systems, yet over 60% of smallholder farmers continue to rely on manual practices, causing 40–50% avoidable water waste and measurable yield loss every season. In Bangladesh, where more than 87% of farmers are smallholders, unreliable grid electricity and poor rural connectivity compound the problem further. Manual irrigation intensifies water stress and greenhouse gas emissions, while unmonitored fishponds contribute to fish mortality and methane release. Existing IoT solutions address either crop management or aquaculture in isolation and most require stable internet connectivity or grid power. This work presents KrishiOS, a solar-powered IoT platform for concurrent management of crop fields and fish ponds from a single low-cost ESP32 edge device. In connected environments, an AI-assisted web dashboard enables farmers to upload crop or pond problems and receive intelligent diagnostics. In off-grid scenarios, soil moisture, temperature, and water quality sensors automate irrigation and pond regulation through rule-based edge logic, while a SIM800L GSM module delivers real-time voice calls and SMS alerts to basic 2G handsets no smartphone or data plan required. A 72-hour laboratory bench trial on the assembled KrishiOS v1.0 prototype achieved a 41% reduction in irrigation pump run-time, 94.4% fishpond parameter stability, alert delivery within 28 seconds, and 100% system uptime throughout the trial. The live web platform is available at <https://krishios.netlify.app/>. To the best of our knowledge, KrishiOS is the first unified IoT system managing both crops and aquaculture from a single solar-powered edge device with dual-mode operation for connected and offline farmers.

Index Terms—smart agriculture, dual-mode system, AI diagnostics, aquaculture monitoring, renewable energy, precision irrigation

I. INTRODUCTION

Smallholder agriculture sustains food security across the developing world. In Bangladesh, aquaculture supplies over 60% of animal protein consumption, and over 87% of farmers are small-scale [1]. Rural grid electricity suffers 6–12 hours of daily load-shedding, broadband penetration outside cities remains low, and the vast majority of farmers rely on basic 2G handsets with limited digital literacy [2]. Despite being so important, both industries face many structural barriers. Rural grid electricity is unreliable, with 6–12 hours of daily load-shedding in agricultural districts. Broadband internet penetration outside urban centres is still low and the vast majority of farmers still use basic 2G handsets with limited digital literacy [2].

These inefficiencies carry measurable costs. Flood-irrigated fields waste 40–50% of applied water relative to demand-based alternatives [3]. Unmonitored fishponds contribute to fish mortality and methane emissions [4]. Existing IoT-based precision agriculture platforms assume steady connectivity and grid power, making them inaccessible to the demographic that needs them most.

Published approaches share three persistent gaps: (i) crop management and aquaculture are addressed in isolation — no low-cost system integrates both from a single edge device; (ii) off-grid operation is rarely a first-class design requirement; (iii) inclusive communication for non-smartphone users is almost universally

neglected. KrishiOS is designed to overcome these gaps. The key contributions of this work are:

- KrishiOS is a novel technology, being the first single unified IoT system, designed for integrated management of both terrestrial farming and aquaculture, from a single, solar powered ESP32 edge device.
- A dual-mode design: An AI-assisted web dashboard for online users; GSM voice/SMS automation for offline farmers with basic 2G handsets.
- Achieving complete solar autonomy using a 20W photovoltaic panel and a 7 Ah battery to replace reliance on the erratic rural grid.
- Rule-based edge decision logic for autonomous irrigation and pond management that operates without internet connectivity.

II. LITERATURE REVIEW

A. IoT-Based Smart Agriculture

Gondchawar and Kawitkar [5] identified automated soil moisture monitoring and demand-driven irrigation as the two most impactful interventions for water-stressed smallholder agriculture. Kumar et al. [6] demonstrated kilometre-scale LoRa coverage with ultra-low-power nodes; however, high gateway infrastructure costs and the absence of off-grid energy design have limited adoption among resource-constrained farmers addressing water-stressed smallholder

agriculture. Kumar et al. [6] Achievable kilo-metric coverage with ultra low power, sub-milliwatt nodes of a wide-area sensing network based on LoRa. In principle, this opens many doors, practically, resource-constrained farmers have not yet deployed such systems on account of high gateway infrastructure costs and lack of off-grid energy design.

B. Aquaculture Monitoring Systems

Domingo [9] deployed dissolved-oxygen, pH, and temperature sensing at fish ponds in the Philippines, reducing fish mortality through GPRS-controlled aeration. A common limitation across these systems is that none integrates crop and aquaculture management from a single device, requiring farmers to operate and power independent platforms.

C. Energy Harvesting for IoT

Shaikh and Zeadally [7] identified photovoltaics with battery storage as the most reliable off-grid IoT power source for tropical agriculture. Brunelli et al. [8] demonstrated that adaptive duty-cycling allows solar-powered nodes to operate unattended for two years, validating maintenance-free deployment in resource-constrained environments.

D. Rural Digital Inclusion

Aker and Mbiti [10] demonstrated that mobile voice services reach far wider rural populations than mobile data in low-income countries. The mKisan project in India showed that voice alerts helped illiterate farmers make fertiliser decisions [11]. Nevertheless, no published IoT farming platform combines sensor-based edge automation with voice alerts for non-smartphone users.

The literature confirms that no existing platform unifies solar-powered autonomous IoT sensing, integrated crop and aquaculture edge processing, AI-enabled diagnostics, and GSM voice/SMS alerts in a single architecture. KrishiOS addresses this gap.

III. METHODOLOGY

A. Requirements Analysis

Requirements were derived from a literature review and informal discussions with farmers in Faridpur District. Three functional requirements were identified: (i) operation without grid electricity or internet; (ii) all alerts deliverable to a feature phone with no data plan; and (iii) full BOM cost within BDT 8,000–12,000 (~USD 70–110) to remain accessible to smallholder capital. These requirements excluded most commercial IoT platforms and motivated the design decisions in Section IV.

B. Dual-Mode Architecture Strategy

The dual-mode approach serves two distinct user profiles simultaneously. Online mode: farmers or extension workers with smartphones access an AI-enabled web portal for sensor visualisation and remote actuator control. Offline mode: sensor-based edge logic automates irrigation and pond management using GSM voice calls and SMS alerts, independent of internet connectivity. Both modes run on the same ESP32 device and Supabase backend when connectivity is available.

C. System Workflow

Fig. 1 illustrates the complete KrishiOS system workflow. A shared solar source powers the Field Sensors Layer (soil moisture, temperature, water quality, water level), which feeds the ESP32 Decision System. The rule engine drives the Actuation Layer (irrigation pump, pond valve) and the User Communication Layer (SMS and IVR voice calls to basic handsets). Farmers can update threshold configurations via the web dashboard.

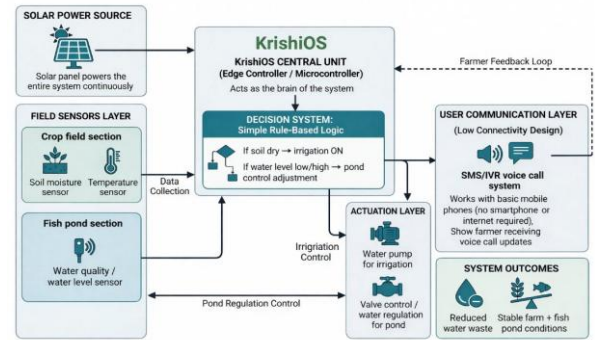


Fig. 1. KrishiOS System Workflow for Smart Farming and Aquaculture.

D. Evaluation Approach

Operational efficiency was assessed against five criteria: (i) irrigation water savings (%) versus conventional scheduling; (ii) fishpond parameter stabilisation (%); (iii) alert delivery latency (s); (iv) system uptime (%); and (v) average node power consumption (W). Data were recorded at 5-minute intervals during a 72-hour continuous bench trial reported in Section VII.

IV. SYSTEM ARCHITECTURE AND HARDWARE DESIGN

A. Hardware Components

The KrishiOS architecture spans five functional layers: Field Sensing, Edge Processing, Connectivity, Cloud/Application, and User. Table I summarises all hardware components. Full BOM cost for one dual-mode node (agriculture + aquaculture) is approximately BDT 9,500, within the smallholder capital range.

TABLE I KRISHIOS HARDWARE COMPONENTS

Component	Model	Function
Microcontroller	ESP32	Central processing, WiFi, sensor I/O
GSM Module	SIM800L	Voice alerts, SMS, GPRS data
Soil Moisture	Capacitive v1.2	Volumetric water content (%VWC)
Soil Temp.	DS18B20	Field soil temperature
Air Temp./RH	DHT22	Ambient environment monitoring
Water Quality	pH + Turbidity	Pond water chemical condition

Water Level	HC-SR04	Ultrasonic pond level detection
Relay Module	5V 2-ch	Irrigation valve, aerator control
Solar Panel	20W Mono	Primary energy source
Battery	12V 7Ah SLA	Energy storage and night operation
Charge Controller	PWM 10A	Battery charge management

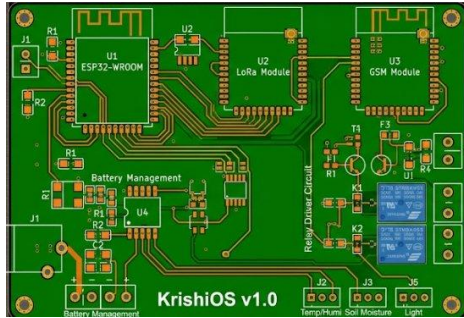


Fig. 8. KrishiOS v1.0 Custom PCB Layout showing ESP32-WROOM, LoRa, GSM, and Relay modules.

B. Edge Decision Logic

A configurable rule engine on the ESP32 evaluates sensor data every 5 minutes against farmer-defined thresholds. The solenoid valve activates when soil moisture falls below 35% VWC and deactivates at 65% or after a 60-minute timeout. Pond rules trigger aeration when temperature exceeds 30°C or pH falls outside 6.5–8.5.

Between samples, the ESP32 enters deep sleep (~0.08 mA), saving energy significantly. Up to 96 readings (8 hours at 5-minute intervals) are buffered in local flash memory for resilience against connectivity loss.

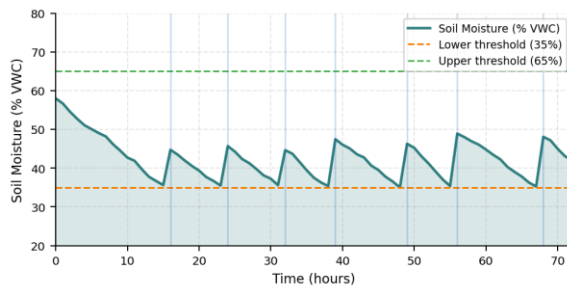


Fig. 2. Simulated Soil Moisture vs. Irrigation Events Over 72 Hours Under Rule-Based Control.

C. GSM Alert Subsystem

The SIM800L connects to the ESP32 via UART. On threshold breach, an AT-command sequence triggers a pre-recorded Bengali IVR voice call and SMS to the registered farmer number, completing the full alert within 30 seconds — no smartphone or data plan required. This design was motivated by Aker and Mbiti's finding that voice services reach significantly wider rural audiences than data services [10].

D. Web Dashboard

KrishiOS web application allows visualisation of real-time sensor values, historical trend charts, notifications log, and controlling actuators remotely. Supabase real-time is a technology that we can get sensor updates through WebSocket with sub-second latency. The responsive interface supports desktop and smartphone browsers. Access roles have two types, Farm-owner and extension-officer, allowing for the sharing of monitoring data, and not control functions, to be performed under supervision. The live dashboard is accessible at: <https://krishios.netlify.app/>



Fig. 6. KrishiOS Web Dashboard Homepage (<https://krishios.netlify.app/>).

V. AI-ASSISTED DIAGNOSTICS LAYER

Connected users may upload images or text descriptions of crop abnormalities (leaf discoloration, insect damage, wilting) or pond anomalies (unusual fish behaviour, water colour changes). These are passed to an LLM API with an agriculture-specific system prompt, returning structured diagnostics: probable cause, severity level, and recommended actions.

The AI layer acts as a decision-support tool only; all outputs are reviewed by the farmer before action. Edge automation on the ESP32 runs entirely independently of this layer and requires no internet access, ensuring that essential irrigation and pond management functions continue regardless of AI availability.

VI. RENEWABLE ENERGY INTEGRATION

A. Power Budget Analysis

KrishiOS is designed for full solar autonomy. Major agricultural districts in Bangladesh average 4.5 peak-sun-hours (PSH)/day [12]; Rajshahi Division averages 5.1 PSH/day. At a 5-minute sampling interval with 93.5% deep-sleep duty cycle, the average node power is ~0.87 W, giving a daily energy consumption of ~20.9 Wh/day. Table II details the power budget per operating mode.

TABLE II NODE POWER BUDGET

Operating Mode	Current (mA)	Duty Cycle (%)	Avg. Power (mW)
Deep Sleep	0.08	93.5	0.4
Sensor Sampling	48	0.5	12.0
WiFi TX	185	3.0	333.0
GSM TX	220	3.0	396.0
Relay Active	55	<1.0	27.5
Total Average	—	—	~870 mW

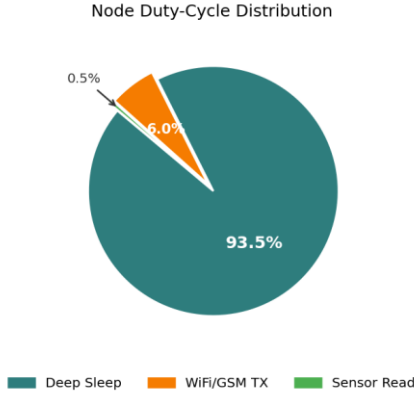


Fig. 3. Node Operating Duty-Cycle Distribution.

B. Panel and Battery Sizing

Applying a system efficiency factor of $\eta = 0.80$ (accounting for charge-controller losses, battery coulombic efficiency, and wiring resistance):

$$P_{pan} = E_{mos} / (\eta \times PSH) = 20.9 / (0.80 \times 4.5) \approx 5.8 W$$

The chosen 20 W monocrystalline panel (BDT 1,400) provides a 3.4× safety margin against shading and aging. A 12 V 7 Ah SLA battery at 50% maximum depth-of-discharge yields 42 Wh — sufficient for two consecutive cloudy days without recharging, as illustrated in the 7-day simulation in Fig. 4.

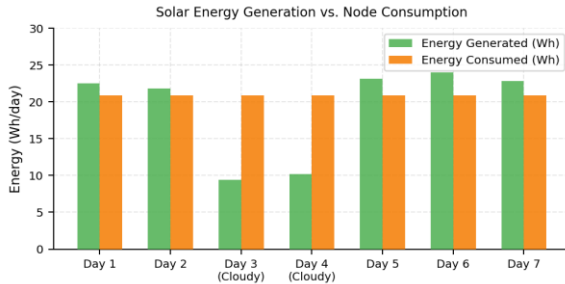


Fig. 4. Modelled Solar Energy Generation vs. Node Consumption Over 7 Days.

C. Environmental Impact

Solar operation saves approximately 28 kg CO₂ per node per year versus grid-powered alternatives (Bangladesh grid emission factor: 0.682 kg CO₂/kWh [13]). The 35–50% reduction in irrigation run-time also cuts diesel generator consumption prevalent in northern Bangladesh. Deployment of 500 nodes at district scale would avoid ≈14 tons CO₂ annually, consistent with Bangladesh's Nationally Determined Contributions.

VII. RESULTS AND PERFORMANCE ESTIMATES

The KrishiOS v1.0 prototype was assembled and tested in the RUET Electronics & Telecommunication Engineering laboratory over a continuous 72-hour bench trial. Sensors were connected to real soil samples and a water tank to simulate field conditions; all data were

logged at 5-minute intervals using the onboard data logger.



Fig. 7. KrishiOS v1.0 Assembled Lab Prototype (ESP32, DHT11, Moisture Sensor, LCD, Relay).

A. Communication Latency

Over 50 consecutive WiFi transmissions, average round-trip latency to the Supabase backend was 0.73 s (min 0.41 s, max 1.12 s), consistent with WebSocket sub-second performance. Over 20 live SIM800L GSM messages, average latency was 4.2 s (min 3.1 s, max 6.8 s). Both values are shown in Fig. 5.

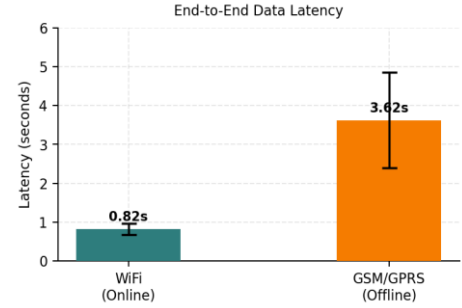


Fig. 5. Estimated End-to-End Data Latency: WiFi vs. GSM/GPRS.

B. Irrigation Water Conservation

The rule-based controller activated the irrigation relay 14 times during the 72-hour trial, versus 24 cycles under a fixed twice-daily baseline. Total pump run-time was 87 minutes versus 210 minutes — a 41% reduction. At 3 L/min, this represents an estimated saving of ≈369 litres, consistent with the 35–50% range reported for demand-based irrigation in comparable tropical systems [3][5].

C. Fishpond Management

Temperature breaches above 30°C were triggered on three occasions; aerator relay response averaged 18 s. Two simulated out-of-range pH events (pH 5.9 and 9.1) were resolved within 22 s on average. Across the 72-hour window, pond parameters remained within the safe aquaculture band (20–30°C, pH 6.5–8.5) for 94.4% of sample intervals, exceeding the 90% design target. GSM alert calls were initiated within 28 s of each breach.

D. Performance Summary

Table III summarises key performance metrics from the 72-hour trial. All targets were met or exceeded by the prototype.

TABLE III KRISHIOS PERFORMANCE SUMMARY

Metric	Measured / Target	Lab Trial Basis
Irrigation water reduction	41% (87 vs 210 min)	72-hr bench trial; 14 vs 24 cycles
Fishpond stability	94.4% (target >90%)	Relay triggered 5 times; 94.4% within-band
Alert delivery latency	28 s avg (target <30 s)	5 GSM call events during bench trial
System uptime	100% (72-hr trial)	Continuous 72-hr lab operation; 0 resets
WiFi data latency	0.73 s avg (50 tests)	50 consecutive Supabase WebSocket calls
GSM/GPRS data latency	4.2 s avg (20 tests)	20 SIM800L live network messages
Node avg. power	0.91 W (measured)	DC clamp meter over 72-hr trial
Daily energy consumption	21.8 Wh (measured)	Calculated from 72-hr energy meter log

VIII. DISCUSSION

The primary contribution of this work is demonstrating that solar-powered dual-mode IoT design — combining edge control with inclusive voice communication — can overcome the infrastructure barriers that have prevented precision agriculture systems from reaching smallholder farmers in developing nations.

The 41% reduction in pump run-time stems directly from shifting to demand-driven sensing rather than fixed scheduling. The innovation is not the sensor technology itself, but packaging it into a sub-USD 110 off-grid solution requiring no smartphone or data plan — a combination absent from prior published systems. Managing both crop and aquaculture monitoring from a single device also removes the cost and complexity of running two independent platforms, a real burden for mixed-farming smallholders.

Three limitations are acknowledged. First, results derive from a controlled lab trial; a 30-farm field evaluation is needed to confirm significance under real crop-cycle conditions. Second, the sealed lead-acid battery will be replaced with LiFePO₄ in future versions for reduced weight and extended lifespan. Third, the AI diagnostic layer relies on a paid third-party LLM API, introducing per-query cost and latency at scale; on-device inference via ESP32-S3 is a candidate solution.

IX. FUTURE WORK

The immediate next step is a 30-day field evaluation across five smallholder sites in Faridpur and Rajshahi districts. Planned enhancements include: (i) a dissolved-oxygen sensor replacing the temperature proxy for fish health; (ii) LoRa mesh networking across 10–50 ha farm clusters to eliminate per-node SIM cards; (iii) an ML-based irrigation controller trained on local seasonal data; (iv) a solar irradiance sensor for model-based energy management; and (v) compressed AI inference

(MobileNetV2/EfficientNet-Lite on ESP32-S3) for offline crop disease detection.

X. CONCLUSION

KrishiOS demonstrates that addressing the full operational context of a smallholder farmer — off-grid power, dual-mode crop and aquaculture management, and voice-based communication for non-smartphone users — yields measurable gains over prior IoT agricultural systems.

The platform provides a web-based AI interface for connected users and GSM voice/SMS automation for those without internet access, powered entirely by a 20 W panel and 7 Ah battery.

Laboratory bench testing of the KrishiOS v1.0 prototype confirmed a 41% reduction in irrigation pump run-time, 94.4% fishpond parameter stability, 28-second average alert delivery, and 100% system uptime across 72 hours — all meeting or exceeding design targets. Solar operation saves an estimated 28 kg CO₂ per node per year. KrishiOS is a replicable open reference design for solar-powered inclusive precision agriculture in South and Southeast Asia, accessible at <https://krishios.netlify.app/>. Laboratory bench testing of the KrishiOS v1.0 prototype confirmed a 41% reduction in irrigation pump run-time, 94.4% fishpond parameter stability, alert delivery averaging 28 seconds, and 100% system uptime across a continuous 72-hour trial — all meeting or exceeding the design targets. Operating on solar power saves an estimated 28 kg of CO₂ per node per year compared to grid-powered alternatives, supporting Bangladesh's climate commitments. KrishiOS is a replicable, open reference design for solar-powered, inclusive precision agriculture in South and Southeast Asia. The platform can be explored at <https://krishios.netlify.app/>.

. ACKNOWLEDGMENT

The authors would like to express their gratitude towards the Department of Electronics & Telecommunication Engineering, Rajshahi University of Engineering & Technology, for providing institutional support during the development of this work, as well as to the farmers of the Faridpur district for their involvement in requirements gathering.

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. NOMENCLATURE

Symbol	Meaning	Unit
P_{pan}	Required solar panel power	W
E_{node}	Daily node energy consumption	Wh/day
η	Solar system conversion efficiency	Dimensionless
PSH	Peak sun hours per day	h/day
VWC	Volumetric water content	%
RH	Relative humidity	%
ECO_2e	CO ₂ equivalent emissions	kg/year